



CHALLENGE 20 / 20

THE BLACKOUT BRIEFCASE

EVIDENCE SURVIVABILITY AND INTEGRITY AFTER ABRUPT POWER LOSS

NATIONAL LEVEL HACKATHON • 2026

POWERED BY SYNTHOQUEST PVT LTD

Student Project Development Challenge • No organizer-provided evidence package.

1 | PROBLEM BACKGROUND

At 02:14 a regional headquarters of an infrastructure company suspected an intrusion. At 02:31 the building lost power and the backup supply failed to carry it. Two server racks lost power mid-transaction, storage controller caches were lost, hypervisors crashed and databases needed recovery. The head of operations' briefcase held a laptop, phone, hardware token and external SSD, all later found powered off at unknown times. The laptop is fully encrypted, and the SSD held an in-flight copy.

The grid operator calls it a grid event. The power-supply maintenance vendor's logs show a remote configuration change two days earlier. Staff then ran filesystem repair and database recovery, which themselves rewrote evidence. Torn pages, replayed journals and partial writes are everywhere.

Nobody knows what was lost, what was repaired, and what was fabricated by the repair.

2 | PROBLEM STATEMENT

Develop an **Evidence Survivability and Integrity Assessment System** that models what each interruption and each later recovery action did to the evidence, grades every recovered item as intact, repaired, reconstructed, corrupted or lost, and records the provenance of each repair.

The system must infer pre-interruption state where it can, test whether the pattern of loss across independent systems is consistent with a coincidental outage or a coordinated one, and state which investigative conclusions remain valid.

Hard constraints

- Well-meaning recovery work altered evidence, and repaired data must never be treated as original.
- Torn writes and in-flight operations leave states that never existed.
- Several independent systems and encrypted devices must be assessed together.
- Coordination of the outage across systems must be tested as a hypothesis, not assumed.
- The set of surviving conclusions must be computed, not asserted.

No evidence package or dataset is provided. Teams decide what to build, what data they need, how to simulate the environment and how to validate the result. No tools or architecture are prescribed.

3 | MANDATORY FEATURES

Teams must develop a working software solution. The system must provide the following features. How each is achieved is for the team to decide.

01	Interruption Effect Modelling	Ability to model what power loss does to each type of storage.
02	Recovery-Action Impact Analysis	Ability to model what later repair did to evidence.
03	Evidence Reliability Grading	Ability to grade every recovered item and its repair provenance.
04	Pre-Interruption State Inference	Ability to infer state before the interruption.
05	Torn-State Detection	Ability to detect states that never existed.
06	Coordinated-Outage Testing	Ability to test coordination across independent systems.
07	Surviving-Conclusion Computation	Ability to compute which conclusions remain valid.
08	Encrypted-Device Reasoning	Ability to reason about evidence on devices that cannot be read.

The system must clearly distinguish between:

RELIABLE	DAMAGED	PARTIALLY RECOVERED	LOST	UNKNOWN
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Damaged evidence must not be used as if it were intact.

Build, don't search. This is a development challenge. Teams must create the system and the controlled environment that proves it works. Submitting an analysis of a given dataset is not a solution.

4 | EXPECTED OUTPUT

Live demonstration. The team must show that its system can handle a complex controlled scenario created by the team. The demonstration must show:

01	Teams' own simulated multi-system environment with a controlled power loss
02	Staff recovery actions applied afterwards
03	Grading of every recovered item
04	Inferred pre-interruption state
05	Detection of torn states
06	Test of coincidental versus coordinated outage
07	Conclusions that survive the damage

The team must be able to explain why the system reached each major conclusion.

Validation standard (every team)

- Build a scenario generator with known ground truth and evaluate on at least ten distinct scenarios, including ones with deliberately misleading evidence.
- At the demonstration, judges change scenario parameters on the spot. The system must adapt without being rebuilt.
- Include an adversary variant that actively tries to defeat the system, and report how the system held up.
- Stated confidence must match measured accuracy. Overconfident or unexplained conclusions lose marks.
- No real systems, real personal data or functional malware may be used.

Submission. Each team must submit:

01	Working Prototype	A functional implementation of the proposed solution.
02	Source Code	Complete source code required to reproduce the project.
03	Technical Architecture	Architecture and component description.
04	Validation Environment	A controlled environment created by the team to demonstrate the solution.
05	Technical Documentation	Implementation, design and testing documentation.
06	Forensic Investigation Report	A professional report generated from the demonstrated investigation.
07	Demonstration	Live technical demonstration to the judging panel.
08	Limitations	Known limitations and situations where the system cannot establish a reliable conclusion.